

Table of contents

Potential as “intrinsic energy of inertia”	2
Electro-Magnetic Field (force caused by potential)	2
Electro-Magnetic interactions/coupling (Maxwell conditions)	4
Potential derivation	4
elec. Potential structure:	4
magn. Potential structure:	5
Electric endogen energy	5
Electric induced energy (exogenous excitation)	5
Power:	6
Frequency:	6
Phase:	6
Phaseshift:	6
Refraction:	6
Dispersion:	6
Dielectricity:	6
Permeability:	6
Conductivity:	7
Conductor (charge conductive element for energy transfer) condition:	7
Slow changing current (quasi-stationary, low frequencies) condition:	7
Energy Balance	7
Mechanical	7
Electromagnetic energy transfer	7
Thermal	8
Relativistic	8
Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potential difference, tension):	8
Low Frequency	9
High Frequency	9
Continuity at Rest: Electric Charge	9
Continuity at Movement: Magnetic Field	9
Continuity at Acceleration: Radiation Energy	9
Continuity and periodic (harmonious oscillation) Movement in VACUUM	10
Continuity and periodic (harmonious oscillation) Movement in MATTER	10
Near Field	10
Far Field	11
Dissipation in Vacuum	11
Dissipation in Matter	11
Energy gain by excitation:	11
Operators	12

Potential as “intrinsic energy of inertia”

Electro-Magnetic Field (force caused by potential)

- static: localized source charge at rest, only spatial dependency, no time dependency
- stationary: source charge at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistic: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movement
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := \text{rot}(A) \stackrel{\text{Matter}}{=} \mu_0 (\mathcal{H} + \mathcal{M}) \stackrel{\text{Vacuum}}{=} \frac{1}{c} \left(\frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left(c p_{\text{dipol}} + m_0 \right) \quad (1)$$

Magnetic Field in Matter

Excitation $\hat{=}$ Displacement

$$\text{Excitation:} \quad \mathcal{H} = \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} = \frac{1}{\mu} \mathcal{B} \quad \text{Magnetisation:} \quad \mathcal{M} = \frac{\mu - \mu_0}{\mu \mu_0} \mathcal{B} = \frac{\mu - \mu_0}{\mu_0} \mathcal{H} \quad (2)$$

Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservativ}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} \cdot \frac{v}{c}\right)\right]}{r}}_{\text{acceleration}} \right] \quad (3)$$

Electric Field in Matter

Displacement $\hat{=}$ Exitation (field change)

dielectricum = non-conducting material

electricum = conductor medium

$$\text{Exitation:} \quad \mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} = \stackrel{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n \langle \mathcal{E}_m \rangle \quad (4)$$

Abbreviations

Φ = electric potential A = magnetic potential $j = \nabla \times \mathcal{H}$ = elec. current density

$\mathcal{E}$ = Electric Field $\mathcal{B}$ = Magnetic Field $\nabla \cdot \mathcal{B} = 0$ $\mathcal{D}$ = Displacement density of dielectricum $\mathcal{M}$ = magnetization

$\mathcal{P}$ = Polarisation $\mathcal{H}$ = magn. exitation, magn. field strength ("magn. displacement")

n = spacial molecule density in volume

$\alpha = 4\pi r^3 = \text{constant}$ χ = elec. susceptibility of dielectricum $\kappa = \frac{\mu - \mu_0}{\mu_0}$ = magn. susceptibility

$\langle p_m \rangle$ = spacial average of magn. dipol moment $\langle \mathcal{E}_m \rangle$ = spacial average of electric field

$\bar{\mathcal{E}}_m$ = time average of electric field μ = magn. permeability μ_0 = magn. constant ...

ε = dielectricity constant ε_0 = ...

Electro-Magnetic interactions/coupling (Maxwell conditions)

conservativ, static charge localisation:

...

stationary, slow charge movement:

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \quad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} = i\omega \mathcal{B} \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} = j - i\omega \varepsilon_0 \mathcal{E} \quad (5)$$

relativistic, fast charge movement:

...

Potential derivation

⇒ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} U = J \quad (6)$$

⇒ Potential caused by “ignition J” (inhomgenity) from source at rest, $v = 0$:

$$U = -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \quad (7)$$

elec. Potential structure:

$$J = -\frac{1}{\varepsilon_0} \varrho$$

$$U_e := \Phi \stackrel{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi} \frac{1}{\varepsilon_0} \int \frac{\varrho}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} Q = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} It \quad (8)$$

magn. Potential structure:

$$J = -\mu_0 j$$

$$U_m := A \stackrel{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \mu_0 I = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2 \varepsilon_0} I = \frac{1}{c^2} \frac{1}{t} U_e \stackrel{\text{source in uniformal movement, } v=\text{const}}{=} \quad (9)$$

Electric endogen energy

- Bond Energy
- Binding Energy

$$E_{pot} = mc^2$$

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (10)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \quad (11)$$

Electric induced energy (exogenous excitation)

- Ionisation Energy
- Excitation Energy

$$E_{kin} = \frac{1}{2} mv^2 \hat{=} (pc)^2 \hat{=} h\nu$$

E_{kin} Energy density electric:

$$w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2 \quad (12)$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (13)$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2} CU^2 \quad (14)$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2} LI^2 \quad (15)$$

Power:

$$P = VI = (U_1 - U_2) \frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (16)$$

Frequency:

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda \frac{1}{T} \stackrel{vacuum}{=} kc \stackrel{matter}{=} k \frac{c}{\eta} = kc \frac{\tan \beta}{\tan \alpha} \hat{=} kc \left(\mathcal{E}_\beta^{-1} \mathcal{E}_\alpha \right)^{-1} = kc \frac{\varepsilon_\beta}{\varepsilon_\alpha} = kc \sqrt{\frac{\varepsilon_0 \mu_0}{\varepsilon \mu}} \hat{=} \sqrt{\frac{k}{m}} \hat{=} \frac{1}{\sqrt{LC}} \quad (17)$$

Phase:

$$\varphi = k \left(\vec{r} \cdot \vec{k} \right) - \omega t := ks - \omega t = \dots \quad (18)$$

Phaseshift:

$$\Delta\varphi = \Delta k \left(\vec{r} \cdot \vec{\Delta k} \right) - \Delta\omega t := \Delta ks - \Delta\omega t = \dots \quad (19)$$

Refraction:

Refraction $\Leftrightarrow \exists$ Matter; refraction causes dispersion

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{\lambda_0}{\lambda} = c\sqrt{\varepsilon\mu} \hat{=} \mathcal{E}_\beta^{-1} \mathcal{E}_\alpha = \frac{\varepsilon_\alpha}{\varepsilon_\beta} = \dots \quad (20)$$

Dispersion:

Dispersion $\Leftrightarrow \exists$ Matter (dielectric medium or conductor); Dispersion $\neq$ Dissipation $\neq$ Diffusion

$$\dots \quad (21)$$

Dielectricity:

$$\varepsilon(\omega) = \dots \quad (22)$$

Permeability:

$$\mu(\omega) = \dots \quad (23)$$

Conductivity:

$$\sigma(\omega) = \dots \quad (24)$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \quad (25)$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (26)$$

Energy Balance

Mechanical

...

Electromagnetic energy transfer

charge conservation:

$$\frac{\partial}{\partial t} \varrho + \nabla \cdot j = 0 \quad (27)$$

$\varrho = \sigma \delta$ = charge density in interior

σ = charge density at border (surface)

δ = Dirac delta-function

$j = \sigma (\mathcal{E} + v \times \mathcal{B}) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t} \mathcal{D} \hat{=}$ electric current density

$j \cdot \mathcal{E} \hat{=}$ energy gain

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$ energy transport density (Intensity)

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (w_e + w_m) + \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (28)$$

Thermal

...

Ralativistic

...

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potential difference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (29)$$

$$\Delta \cdot := \text{div grad} = \text{Laplacian}$$

$$J = \text{"ignition of inhomogeneity, cause"} \quad j = \text{Current density} \quad \varrho = \text{Charge density}$$

$$V = \text{Potential difference} \quad I = \text{Current} \quad R = \text{Resistance}$$

$$L = \text{Inductance} \quad C = \text{Capacitance} \quad Q = \text{charge (ion)} \quad \Psi = LI = \text{mag. Flux}$$

$$\mathcal{S} = \mathcal{E} \times \mathcal{H} = \text{Poynting-Vector} = \text{radiation intensity}$$

$$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr = \text{dissipation energy lost from radiation} \approx 0 \text{ for low frequencies } \omega$$

$$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor} \quad r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"}$$

μ = permeability of conductor σ = conductivity of conductor (dielectricum)

Low Frequency

$$\begin{aligned} \omega &\ll \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \\ L &\approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0 \end{aligned} \tag{30}$$

High Frequency

$$\begin{aligned} \omega &\gg \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \\ L(t) &\neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0 \end{aligned} \tag{31}$$

Continuity at Rest: Electric Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emergence of Radiation

Continuity and periodic (harmoneous oscillation) Movement in VACUUM

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \nu \lambda = \frac{\lambda}{T}$$

$$2\pi = k\lambda = T\omega = \frac{\omega}{\nu}$$

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \quad \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

Continuity and periodic (harmoneous oscillation) Movement in MATTER

$$\text{Refraction index } \eta \text{ emerges: } \varepsilon\mu = \left(\frac{\eta}{v_{ph}}\right)^2 \hat{=} \left(\frac{\eta}{c}\right)^2$$

$$\text{Dispersion: } k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$$

Near Field

granular spacetime

$$r \ll \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\varepsilon_0 r^6}}{\frac{k^2 \mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$$\vec{p} = \text{elec. Dipol}$$

Far Field

flat spacetime

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|} \right)^2}{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|} \right)^2} = 1$$

Dissipation in Vacuum

no Continuity of charge/matter, no Refraction index, Reflection

Dissipation in Matter

no Continuity of charge/matter with Dielectricum, Refraction merges, Reflection, Absorption

Energy gain by excitation:

Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

$$h = \tan(\alpha) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \quad (32)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (33)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (34)$$

¹Wirkungsquantum

$V_0 =$ Opposing Potential ($I = 0$)

$\alpha = \angle(E_{kin}, \nu)$ $e =$ electron charge $Q =$ ion charge

$C =$ Capacitance $\nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{C}{L} R^2}$

$\Rightarrow$ Inductance $L = \frac{4\pi^2}{C} \left(\frac{e}{h} Q + \nu_0 \right)^2$

$\Rightarrow$ Ionisation Energy of chemical element $E_i = h\nu = h \frac{\lambda}{c} = \hbar\omega = \frac{\hbar}{k\lambda}\omega$

$\Rightarrow$ Energy released (gained) $E_{gain} = E_{pot} - E_i = mc^2 - h\nu$

Operators

Operators

$\nabla := grad =$ Gradient (Gefälle, Neigung)

$\nabla \cdot := div =$ Divergence (Quelle)

$\nabla \times := rot =$ Rotation (Wirbelung, Curl)

$\nabla^2 =$ Lagrange Operator (Krümmung, Beschleunigung)

$\Delta \cdot := div(grad) =$ Laplace Operator

$\Delta :=$ Difference, shift